

FCGUB ②

Es ①

$$① \binom{25}{2} \binom{2}{1} = 25 \cdot 24 = 600$$

$$② \binom{1}{1} \binom{29}{2} \cdot 2 = \frac{29!}{1! \cdot 28!} \cdot 2 = 29 \cdot 2 = 48$$

$\downarrow$
 fra tutti i me
 ne sceglie 2
 2 corde

$$P(A) = \frac{48}{600} = 0,08$$

Es ②

Bonole

Es ④

$$1) \binom{13}{10}$$

$$2) \binom{4}{2} \binom{11}{8}$$

$$3) \binom{4}{1} \binom{11}{9}$$

Es ⑤

$$\uparrow D = \{ \text{Disp. di 7 piazze su 7 SOARELLI} \}$$

$$|D| = 7!$$

$$A = \{ \text{ALFREDO e BRANCA vicini} \}$$

$$|A| = \underbrace{\binom{4}{1} \binom{1}{1}}_{\text{PIAZZA ALFREDO e BRANCA}} \cdot \underbrace{6 \cdot (5!)}_{\text{6 pos.}} \xrightarrow{\text{PIAZZA 6' olt}} = 2 \cdot 6 \cdot 5!$$

$$P(A) = \frac{12 \cdot 5!}{7!}$$

$$2) B = \{ \text{Asfido e Birco um su tuck tuck} \}$$

$$|B| = \binom{4}{1} \binom{1}{1} = 7 \cdot 1!$$

6
6

$$D = \{ \text{contingui di 5 carte to 52} \}$$

$$|D| = \binom{52}{5}$$

1) Pda

$$A = \{ 4 \text{ carte stess colore a 1 carta} \}$$

$$|A| = \binom{4}{4} \cdot \binom{49}{1} \cdot 13 \text{ colori}$$

$$P(A) = \frac{\binom{4}{4} \cdot \binom{49}{1} \cdot 13}{\binom{52}{5}}$$

2) Cda

$$B = \{ 5 \text{ carte stess seme} \}$$

$$|B| = \binom{13}{5} \cdot 4$$

$$P(B) = \frac{\binom{13}{5} \cdot 4}{\binom{52}{5}}$$

3) Fd

$$C = \{ \text{Fis e Coppa} \}$$

$$|C| = \binom{4}{2} \cdot 13 \cdot 12 \cdot \binom{4}{3} \quad P(C) =$$

Folge ③

Ü ①

$$1) \binom{9}{3} \binom{6}{3} \binom{3}{3} \cdot 3$$

$$2) 6!$$

$$3) \frac{\binom{9}{3} \binom{6}{3} \binom{3}{3}}{3!}$$

$$4) \frac{\binom{5}{3} \binom{2}{2} \binom{1}{1}}{2!}$$

$$5) 1$$

Ü ②

$$\mathcal{X} = \{ (w_1, \dots, w_n) \mid w_i = 1 \dots 40 \}$$

$$|\mathcal{X}| = \binom{40}{10}$$

$$1) \binom{1}{1} \binom{1}{1} \binom{1}{1} \binom{37}{2}$$

$$P(A) = \frac{\binom{37}{2}}{\binom{40}{10}}$$

$$2) \binom{34}{4}$$

$$P(B) = \frac{\binom{34}{4}}{\binom{40}{10}}$$

$$\frac{4}{40} \cdot \frac{4}{39} \cdot \frac{4}{38} \cdot \frac{4}{37} \cdot 4!$$

Ü ③

E5 ⑤

$$\begin{aligned}
 1) P(\text{diff}) &= P(A) \cdot P(\text{diff}|A) + P(B) \cdot P(\text{diff}|B) + P(C) \cdot P(\text{diff}|C) \\
 &= P(A) \cdot \frac{P(\text{diff} \cap A)}{P(A)} + \dots + = 0.4 \cdot 0.02 + 0.1 \cdot 0.03 + 0.5 \cdot 0.04 \\
 &\quad \downarrow \\
 &\quad \frac{P(\text{diff}) \cdot P(A)}{P(A)} \quad \swarrow \\
 &= 0.031
 \end{aligned}$$

$$2) P(A|\text{diff}) = \frac{P(A) \cdot P(\text{diff}|A)}{P(A) \cdot P(\text{diff}|A) + P(B) \cdot P(\text{diff}|B) + P(C) \cdot P(\text{diff}|C)} = \frac{0.4 \cdot 0.02}{0.031}$$

$$P(B|\text{diff}) = \frac{P(B) \cdot P(\text{diff}|B)}{P(A) \cdot P(\text{diff}|A) + P(B) \cdot P(\text{diff}|B) + P(C) \cdot P(\text{diff}|C)} = \frac{0.1 \cdot 0.03}{0.031}$$

$$P(C|\text{diff}) = \frac{P(C) \cdot P(\text{diff}|C)}{P(A) \cdot P(\text{diff}|A) + P(B) \cdot P(\text{diff}|B) + P(C) \cdot P(\text{diff}|C)} = \frac{0.5 \cdot 0.04}{0.031}$$

E5 ⑥

$$\begin{aligned}
 1) P(\text{vic. 1}) &= P(\text{inv. 1}) \cdot P(\text{vic. 1}|\text{inv. 1}) + P(\text{inv. 0}) \cdot P(\text{vic. 1}|\text{inv. 0}) \\
 &= 0.55 \cdot \frac{P(\text{vic. 1} \cap \text{inv. 1})}{P(\text{inv. 1})} + 0.45 \cdot \frac{P(\text{vic. 1} \cap \text{inv. 0})}{P(\text{inv. 0})} \\
 &= 0.55 \cdot 0.91 + 0.45 \cdot 0.06 = 0.5275
 \end{aligned}$$

$$2) 0.45 \cdot 0.94 + 0.55 \cdot 0.08 = 0.4725$$

$$3) P(\text{nu } 1 | \text{ric } 1) = \frac{P(\text{nu } 1 \cap \text{ric } 1)}{P(\text{ric } 1)} = \frac{0.55 \cdot 0.81}{0.5225} = 0.84$$

$$4) P(\text{nu } 0 | \text{ric } 0) = \frac{P(\text{nu } 0 \cap \text{ric } 0)}{P(\text{ric } 0)} = \frac{0.65 \cdot 0.94}{0.4725} = 0.89$$

8.3

$$1) P(\text{una alla } S) = P(\text{prob proba 4 e una } S) = (1-p)^4 \cdot p = \left(\frac{19}{32}\right)^4 \cdot \frac{13}{32} \approx 0.934$$

$$2) P(\text{una almeno } 2)$$

$$= \sum_{k=2}^{10} \binom{10}{k} p^k (1-p)^{10-k} = \sum_{k=2}^{10} \binom{10}{k} \cdot \left(\frac{13}{32}\right)^k \cdot \left(\frac{19}{32}\right)^{10-k}$$

8.1

$$\begin{aligned} 1) P(\text{sterzo sex}) &= P(\text{ano}) + P(\text{etero}) \cdot P(\text{sterzo sex} | \text{etero}) \\ &= p + (1-p) \cdot 0,5 = p + \frac{1}{2} - \frac{1}{2}p = \frac{1}{2}p + \frac{1}{2} \end{aligned}$$

$$P(\text{ano} | \text{sterzo sex}) = \frac{P(\text{ano} \cap \text{sterzo sex})}{P(\text{sterzo sex})} = \frac{p}{\frac{1}{2}p + \frac{1}{2}}$$

2)

$$\begin{aligned} P(\text{sterzo dif}) &= P(\text{etero}) \cdot P(\text{sterzo dif} | \text{etero}) = (1-p) \cdot 0,5 = \frac{1}{2} - \frac{1}{2}p \\ &= \frac{P(\text{sterzo dif} \cap \text{etero})}{P(\text{etero})} \end{aligned}$$

8.2

$$\begin{aligned} 1) P(\text{Test}) &= P(TC) \cdot P(T | TC) + P(TT) \cdot P(T | TT) \\ &= \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot 1 = \frac{1}{6} + \frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} = \frac{1}{2} \end{aligned}$$

$$2) P(TC | T) = \frac{P(TC) \cdot P(T | TC)}{P(\text{Test})} = \frac{\frac{1}{3} \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{1}{3}$$

$$3) P(TT | T) = \frac{P(TT) \cdot P(T | TT)}{P(\text{Test})} = \frac{\frac{1}{3} \cdot 1}{\frac{1}{2}} = \frac{2}{3}$$

8.3

$$1) P(\text{voti si} | \text{voti}) = P(v) P(\text{voti si} \cap \text{voti}) = \frac{1}{2} - \frac{1}{2} = \frac{1}{4}$$

2) k si

1ª Parte $\Rightarrow$ k votanti

$$P(k \text{ votanti}) = \sum_{i=k}^m \binom{m}{i} p_v^i (1-p)^{m-i} = \sum_{i=k}^m \binom{m}{i} p_v^i$$

2ª Parte $\Rightarrow$ Tra i m votanti, k si

$$P(k \text{ si } m \text{ votanti}) = \binom{m}{k} p_{si}^k (1-p)^{m-k} = \binom{m}{k} \cdot p_{si}^k$$

$$P(k \text{ si}) = \sum_{i=k}^m \binom{m}{i} \cdot p_v^i \cdot \binom{m}{k} p_{si}^k$$

8.4

1) 200 (201) 67 T

$$\binom{200}{67} p^{67} (1-p)^{133}$$

E_2

$$1) P(X=k) = \frac{1}{6}$$

$$3) VCF = \sum_{i \in (1,5)} (c_i - E(c))^2 \cdot \frac{1}{6} \approx 2,52$$

③

$X = \begin{cases} 1 & \text{so min } (w) = 1 \\ 2 & \text{" " " " } = 2 \\ 3 & \text{" " " } = 3 \\ 4 & \text{" " } = 4 \\ 5 & \text{" } = 5 \\ 6 & \text{" } = 6 \end{cases}$

$$(2,2) \quad (2,3) \quad (2,4) \quad (2,5) \quad (2,6)$$

$$D \quad P(x)$$

$$P(X=1) = \frac{\pi \cdot 1}{\frac{\pi}{2}}$$

$$P(X=2) = \frac{3 \cdot 3 \cdot 1}{36 \cdot 129}$$

$$P(X=3) = \frac{2}{36}$$

$$P(U=q) = \frac{S}{2x}$$

$$P(X=5) = \frac{31}{342}$$

$$P(X=0) = \frac{1}{2}$$

$$P(X=x) = \frac{1}{3} \cdot \frac{(6-x)}{6} + \frac{1}{3}$$

$$E(X) = \sum_{x \in \mathcal{X}} x \cdot P(X=x)$$

Es 5

Alcune volte supero esame

+3 V

-1 se X o no data

$$P(X \text{ corretta}) = \binom{10}{x} p^x (1-p)^{10-x} \Rightarrow X \sim \text{Bin}(n, p) \quad p = \frac{1}{4}$$

$$P(\text{Supero Esame}) = P(S) = P\left(\frac{10}{2}\right) + P\left(\frac{10}{3}\right) + \dots + P\left(\frac{10}{10}\right)$$

Ma non posso prendere tutti i voti in più con il -1 alle domande sbagliate

devo rispondere giusto ad almeno 7 ($7-3-3=10$)

$$P(S) = \sum_{i=7}^{10} \binom{10}{i} p^i (1-p)^{10-i}$$

$$2) E(X) = np = 10 \cdot \frac{1}{4} = 2.5$$

$$V(X) = np(1-p) = 10 \cdot \frac{1}{4} \left(1 - \frac{1}{4}\right) = 10 \cdot \frac{1}{4} \left(\frac{3}{4}\right) = \frac{10}{4} \cdot \frac{3}{4} = \frac{15}{8}$$

Foglio 7

Es 1

DADO EQUO

$$P(\text{5 o 6}) = \frac{1}{3}$$

$$T = \text{n}^\circ \text{ lanci}$$

$$T = \begin{cases} 1 & \text{n}^\circ \text{ lanci} = 1 \\ 2 & \text{n}^\circ \text{ lanci} = 2 \\ \vdots & \\ n & \text{n}^\circ \text{ lanci} = n \end{cases}$$

$$X = \begin{cases} 5 \\ 6 \end{cases}$$

$$1) P(T=3, X=5) =$$

$T=3$ $\Rightarrow$ primi 2 lanci non da 5, al 3° sì

$$T \sim \text{Geom}(p)$$

$$P(T=3, X=5) = (1-p)^2 \frac{1}{6} = \left(\frac{2}{3}\right)^2 \cdot \frac{1}{6} = \frac{2}{27}$$

2) $P(T)$, poiché $T \in \text{Geom}$

$$P(T) = (1-p)^{T-1} p$$

3) $X \in \text{Bin } 2$

$$P(X=x) = \frac{1}{2}$$

Es 2

$$1) P(X_1 + X_2 = z) = \sum_{m=0}^{\min(z,1)} P(X_1=m) \cdot P(X_2=z-m)$$

$$2) E(X_1 + X_2) = E(X_1) + E(X_2)$$

8 ②

$$X \sim \text{Geom}(p) \quad E(X) = \frac{1}{p}$$

$$E(1^{\circ} \text{ loco}) = 1 \quad (\text{quadrare fatto solo una})$$

$$P(\text{non trova foto}) = \frac{1}{2} \Rightarrow P(\text{trova foto diversa}) = \frac{1}{2}$$

$$E(2^{\circ} \text{ loco}) = \frac{1}{\frac{1}{2}} = 2$$

$$P(\text{non trova foto dello zio}) = \frac{2}{3} \Rightarrow P(\text{trova foto diversa}) = \frac{1}{3}$$

$$E(3^{\circ} \text{ loco}) = \frac{1}{\frac{1}{3}} = 3$$

$$E(i\text{-esimo loco}) = \frac{6}{(7-i)}$$

9 ④

- $P_{\text{sub}} (se \leq 200 \text{ e } 1 \text{ si trova tutto})$

$P \sim \text{Geom}(p)$

$$P(\text{trova al ciclo } T) = [(1-p)^T]^k p$$

$$T = T-1$$

$(1-p)^T$ è la prob. che non si trova nei primi T cicli e che viene poi trovato

$$i \text{ k cicli} \Rightarrow [(1-p)^T]^k \cdot p$$

(prob. che si trova)

- $P_{\text{pubb}} =$

8.2

$X = \text{u.a.}$ $n = \text{Anzahl}$ u.a.

Bauk ca $p(3,4)$

$$X_i = \begin{cases} 1 & \text{se } 1 \leq i \leq n \\ 2 & \text{se } n+1 \leq i \leq 2n \\ \vdots & \end{cases}$$

1)

$$P(X=1) = p(1-p) + (1-p) \cdot p$$

$$P(X=2) = p^2(1-p) + (1-p)^2 p$$

$$P(X=3) = p^3(1-p) + (1-p)^3 p$$

:

$$k \text{ Anzahl u.a.} = P(X=k) = p^k(1-p) + (1-p)^k p$$

2)

$$\begin{aligned} E(X) &= \sum_{i=1}^{\infty} i \cdot P(X=i) = \sum_{i=1}^{\infty} i \cdot [p^i(1-p) + (1-p)^i p] \\ &= \sum_{i=1}^{\infty} i \cdot p^i(1-p) + \sum_{i=1}^{\infty} i \cdot (1-p)^i p \end{aligned}$$

$$\Rightarrow V(X) = \sum_{i=1}^{\infty} (i - E(X))^2 P(X=i)$$

$$\Omega = \{ \omega_1, \omega_2, \omega_3 \mid \omega_1 \neq \omega_2, \omega_1 \neq \omega_3, \omega_2 \neq \omega_3 \mid \omega_i = 1, \dots, 40 \}$$

2) P_{mpl}

1 pair 2 3 cost

$$|\Omega| = \binom{40}{3}$$

$$P(\text{mpl}) = 4 \cdot \frac{1}{\binom{40}{3}}$$

$$3 \frac{\binom{40}{3} \cdot 10^3}{\binom{40}{3}} = \frac{4 \cdot 10^3}{\binom{40}{3}}$$

$$P(\omega_i = 10) = \frac{3}{40}$$

$$P(1^{\text{st}} \text{ est } | 10) = P(i=1 \mid \omega_i = 10) = \frac{P(i=1 \cap \omega_i = 10)}{P(\omega_i = 10)}$$

$$= \frac{1 \cdot \frac{1}{40}}{\frac{3}{40}} =$$

$$P(A_{unc} 3) = 0,3 \cdot 0,3 \cdot 0,3$$

$$3! \cdot (0,3 \cdot 0,2 \cdot 0,5)$$

$$Z =$$

Foglio 8

Es 2

2% pop. mondiale mancina

100 individui, quale prob. che 3 siano mancini

$$X \sim \text{Poisson}(\lambda = 2) = \frac{1}{50}$$

$$P(\text{che 3 mancini}) = P(\text{mancini} \geq 3) = 1 - P(\text{mancini} < 3)$$

$$P(\text{mancini} < 3) = P(X < 3) = \sum_{k=0}^2 e^{-\frac{1}{50}} \frac{\left(\frac{1}{50}\right)^k}{k!}$$

Es 3

Y : n° email spam (arrivano con prob = p)

Z : n° email legittime (arrivano con prob = $1-p$)

X : n° email $\sim \text{Poisson}(1)$

Distribuzione Y (arrivano anche Y email spam) $\downarrow$ p poiché è la prob. che
arriva spam

$$P(Y=y) = \sum_{m=0}^{\infty} P(X=m) P(Y=y | X=m) = e^{-p} \frac{(1-p)^y}{y!}$$

$$P(Z=z) = \sum_{m=0}^{\infty} P(X=m) P(Z=z | X=m) = e^{-(1-p)} \frac{(1-p)^z}{z!}$$

8 ②

Dist. margind.

X:

$$P(X=x) = \frac{\binom{3}{x} \binom{4}{3-x}}{\binom{7}{3}} \quad x = 0, 1, 2, 3$$

Y:

$$P(Y=y) = \frac{\binom{2}{y} \binom{5}{2-y}}{\binom{7}{3}} \quad y = 0, 1, 2$$

Conjunta

$$P(X=x, Y=y) = \frac{\binom{3}{x} \binom{2}{y} \binom{2}{3-(x+y)}}{\binom{7}{3}}$$

$$x = 0, 1, 2, 3$$

$$y = 0, 1, 2$$

3)

$$P(\text{Ganzes}) = P(X=3, Y=0) + P(X=2, Y=1) + P(X=1, Y=2)$$

$$= \frac{1}{35} + \frac{\binom{3}{2} \binom{2}{1}}{35} + \frac{\binom{3}{1} \binom{2}{2}}{35} = \frac{3}{35} + \frac{6}{35} + \frac{1}{35} = \frac{10}{35} = \frac{2}{7}$$

$$\binom{7}{3} = \frac{7!}{3! 4!} = \frac{7 \cdot 6 \cdot 5}{3!} = 35$$

Es 6

1 Dado con 2B, 2R, 3V

1 2 3 4 5 6

$$R = \begin{cases} 1 & \omega = 2, 3 \\ 0 & \omega = 1, 4, 5, 6 \end{cases}$$

$$V = \begin{cases} 1 & \omega = 4, 5, 6 \\ 0 & \omega = 1, 2, 3 \end{cases}$$

$V \backslash R$	0	1	
0	$\frac{1}{6}$	0	$\frac{1}{6}$
1	$\frac{3}{6}$	0	$\frac{3}{6}$
	$\frac{4}{6}$	$\frac{0}{6}$	

} Distribuzione R

Na Sono indipendenti

distibuzione V

2)

$$Z = \begin{cases} 1 \\ 0 \end{cases}$$

Es 7

Componente left = p (non right = (1-p))

Contabile = 0 (non contabile = (1-a))

$$^0 P(R_c = k)$$

$$R_i \sim \text{Bin}(m, p)$$

$$R_{ci} = \begin{cases} 1 & \text{if } p = p \\ 0 & \text{if } p = 1-p \end{cases}$$

$$P(R_c = k) = \binom{m}{k} p^k (1-p)^{m-k}$$

Passo due die

$$(F_c = f_c, F_u = f_u, R_c = r_c, R_u = r_u) \sim \text{Multinomial}(m, (1-p)d, (1-p)(1-d), p(1-d), p(1-d))$$

$$\text{h.f.d.} \quad F_c = \begin{cases} 1 & (1-p)d \\ 0 & 1 - (1-p)d \end{cases} \quad F_u = \begin{cases} 1 & (1-p)(1-d) \\ 0 & 1 - (1-p)(1-d) \end{cases} \quad R_u = \begin{cases} 1 & p(1-d) \\ 0 & 1 - p(1-d) \end{cases}$$

$$P(F_c = f_c, F_u = f_u, R_c = r_c, R_u = r_u) = \frac{m!}{f_c! f_u! r_c! r_u!} \cdot ((1-p)d)^{f_c} ((1-p)(1-d))^{f_u} (p(1-d))^{r_c} (1-p(1-d))^{r_u}$$

2)

Calor ca Multinomial

$$(F, R_u, R_c) \sim \text{Multinomial}(m, (1-p), p(1-d), p)$$

$$P(R_u = r_u | R_c = k) = \frac{P(F = m - k - r_u, R_c = k, R_u = r_u)}{P(R_c = k)} \quad \text{calcule prin}$$

$$= \frac{n!}{(n-k-m)! 2m! k!} \cdot ((1-p))^{m-k-m} \cdot (p(1-p))^{2m} \cdot (1-p)^k$$

$$\binom{n}{k} p^k (1-p)^{n-k}$$

Ex ⑧

$$Z = X(1-Y)$$

$$W = 1 - XY$$

Z:

$Y \backslash X$	0	1
0	0	1
1	0	0

W:

$Y \backslash X$	0	1
0	1	1
1	1	0

$Z \backslash W$	0	1
0	p^2	$p(1-p) + (1-p)^2$
1	0	

Fo gli 10

6.2

$$X \sim N(2, 25)$$

$$Z = \frac{X - (2)}{5} \Rightarrow X = 5Z + 2$$

$$1) P(|X - 2| > 7) = P(|5Z - 2 + 2| > 7) = P(|5Z| > 7)$$

$$P\left(Z < -\frac{7}{5}\right) + P\left(Z > \frac{7}{5}\right) = 2P\left(Z > \frac{7}{5}\right) = 2 \cdot [1 - P\left(Z \leq \frac{7}{5}\right)]$$
$$= 2 \cdot [1 - 0,8413] = 0,1616$$

$$2) P(0 \leq X \leq 7) = P(0 \leq 5Z + 2 \leq 7) = P(-2 \leq 5Z \leq 5)$$

$$= P\left(-\frac{2}{5} \leq Z \leq 1\right) = P(Z \leq 1) - P\left(Z \leq -\frac{2}{5}\right)$$

$$= P(Z \leq 1) - [1 - P\left(Z > \frac{2}{5}\right)] = 0,8413 - (1 - 0,6915) = 0,4567$$

$$3) \text{ Trova } d / P(X > d) \leq 0,1$$

$$P(5Z + 2 > d) \leq 0,1 = P(5Z > d - 2) \leq 0,1$$

$$P\left(Z > \frac{d-2}{5}\right) \leq 0,1 \rightarrow P\left(Z \leq \frac{d-2}{5}\right) \geq 0,9$$

$$P\left(Z \leq 1,28\right) \geq 0,9 \quad \frac{d-2}{5} = 1,28 \Rightarrow d = 8,4$$

8) 3

$$X \sim N(5, 0.25)$$

$$Z = \frac{X - 5}{0.5} = 2X - 5 \quad X = \frac{Z}{2} + 5$$

1) Na simplifica: $Q_{int} \Rightarrow$

$$\begin{aligned} P(X < 4) + P(X > 6) &= P\left(\frac{Z}{2} < -1\right) + P\left(\frac{Z}{2} > 1\right) \\ &= P(Z < -2) + P(Z > 2) = 2P(Z > 2) = 2 \cdot [1 - P(Z \leq 2)] \\ &= 2 \cdot [1 - 0.9772] = 0.0456 = 4.56\% \end{aligned}$$

2) Verifico trace σ^2 offmche $\% \text{ var} \geq 1\%$

$$Z = \frac{X - 5}{\sigma} \quad X = \sigma Z + 5$$

$$\begin{aligned} P(X < 4) + P(X > 6) &= P(\sigma Z + 5 < 4) + P(\sigma Z + 5 > 6) \\ &= P(\sigma Z < -1) + P(\sigma Z > 1) \end{aligned}$$

$$= 2P(Z > \frac{1}{\sigma}) = 2 \cdot [1 - P(Z \leq \frac{1}{\sigma})] \leq 0.01$$

$$= [1 - P(Z \leq \frac{1}{\sigma})] \leq 0.005$$

$$= 1 - P(Z \leq \frac{1}{\sigma}) \leq 0.005$$

$$P(Z \leq \frac{1}{\sigma}) \geq 0.995$$

$$\sigma = 0,254 \text{ in } \text{in}^2$$

$$\frac{1}{\sigma} = 3,4 \text{ de nelli}$$

$$\text{Tr. Gasman } 79,555$$

8,4

$$\mu = \begin{cases} -2 & \text{se } \text{intra } 0 \text{ (A)} \\ +2 & \text{se } \text{intra } 1 \text{ (A)} \end{cases}$$

$$X = \mu + Z \quad Z \sim N(0,1)$$

$$X = \begin{cases} 2,5 & B = 1 \\ -2,5 & B = 0 \end{cases}$$

$$1) P(B \text{ decodifica } 1 | A_{\text{mis } 0}) =$$

$$\mu = -2 \quad X = -2 + Z$$

$$= P(X \geq 0,5) = P(Z - 2 \geq 0,5) = P(Z \geq 2,5)$$

$$= 1 - P(Z \leq 2,5)$$

$$= 1 - 0,9938 = 0,0062$$

$$2) P(B \text{ decodifica } 0 | A_{\text{mis } 1}) =$$

$$\mu = 2 \quad X = 2 + Z$$

$$= P(X < 9,5) = P(2+Z < 9,5) = P(Z < 7,5)$$

$$= P(Z < 7,5) = 1 - P(Z < 1,5) = 1 - 0,9332 = 0,0668$$

$$3) P(A_{nu 0}) = P(A_{nu 1}) = \frac{1}{2}$$

$$P(B_{dec 1}) = \overset{n=2}{P(A_{nu 1}) \cdot P(B_{dec 1} | A_{nu 2})} + \overset{n=2}{P(A_{nu 0}) \cdot \underbrace{P(B_{dec 1} | A_{nu 0})}_{0,0062}}$$

$$= \frac{1}{2} \cdot P(B_{dec 1} | A_{nu 1}) + \frac{1}{2} \cdot 0,0062$$

$$= \frac{1}{2} \cdot P(X \geq 9,5) + 0,0031 =$$

$$Z = 2 + X$$

$$= \frac{1}{2} \cdot P(2+Z \geq 9,5) + 0,0031 = \frac{1}{2} \cdot P(Z \geq 7,5) + 0,0031$$

$$= P(Z \geq 7,5) = 1 - P(Z \leq 7,5) = 1 - P(Z \leq 1,5)$$

$$= 1 - [1 - P(Z \leq 1,5)]$$

$$= 1 - (1 - 0,9332) = 0,9332$$

$$= \frac{1}{2} \cdot 0,9332 + 0,0031 = 0,4666 + 0,0031 = 0,4697$$

$$④ P(A_{nu 2} | B_{dec 1}) = \frac{P(A_{nu 2} \cap B_{dec 1})}{P(B_{dec 1})} = \frac{P(B_{dec 1} | A_{nu 2}) \cdot P(A_{nu 2})}{P(B_{dec 1})}$$

$$= 0,5534$$

6 ⑤

$$P(11) = \frac{1}{36} \sim B_n(n, \frac{1}{36}) = B_{11}(36, \frac{1}{36})$$

$$1) E(X) = n \cdot \frac{1}{36} = \frac{36}{36} = \frac{27}{3}$$

$$V(X) = np(1-p) = \frac{36}{36} \cdot (1 - \frac{1}{36}) = \frac{36-1}{36} = \frac{87}{108}$$

$$2) X \sim \text{Gas}(\frac{4}{3}, \frac{87}{108})$$

$$Z = \frac{X - \frac{27}{3}}{\sqrt{\frac{87}{108}}} \Rightarrow X = \frac{10 \cdot \sqrt{108}}{36} Z + \frac{27}{3}$$

$$P(X > 10) = P(Z > \frac{6}{108}) = 1 - P(Z < \frac{6}{108}) = 1 - P(Z < 0,55) =$$

$$1 - 0,7224 = 0,2776$$